**SWINBURNE UNIVERSITY OF TECHNOLOGY**
COS40005 · Computing Technology Project A

# Team and Project Plan

Internal Assistant for Enterprise Knowledge (Slack App Bot)

| Item | Details |
|---|---|
| **Unit code** | COS40005 |
| **Unit name** | Computing Technology Project A |
| **Submission date** | 25 October 2026 |
| **Project sponsor** | Mr. Phat (Eric) Pham |
| **Project manager** | Truong Ngoc Gia Hieu |
| **Release covered** | Alpha (Project A, Weeks 5–12) |
| **Document version** | 1.0 – planning baseline; schedule revised 25 October 2026 |

| Team member | Student ID |
|---|---|
| Truong Ngoc Gia Hieu (project manager) | 105565520 |
| Pham Nguyen Minh Hoang | 105543500 |
| Nguyen Quang Khai | 105560512 |
| Lu Nhat Hoang | 105234956 |
| Duong Nguyen Dang | 105508444 |
| Le Xuan Ny | 105680151 |

# Table of Contents

**Acknowledgment of Country** ..... **4**

**PART 1 Team Code of Conduct** ..... **5**

1. Team profile ..... 5
2. Team role ..... 6
3. Teamwork roadmap ..... 7
   - 3.1 Weekly work pattern and communication protocol ..... 7
   - 3.2 Task estimation, assignment and monitoring ..... 7
   - 3.3 Communication rules, expected behaviour and fallback plans ..... 7
4. Document management ..... 8
5. Risk mitigation ..... 8

**PART 2 Project overview and plan** ..... **11**

1. Problem statement ..... 11
   - 1.1 Background ..... 11
   - 1.2 Problem ..... 11
   - 1.3 Aim, purpose and objectives ..... 11
2. Scope ..... 12
   - 2.1 In scope ..... 12
   - 2.2 Out of scope ..... 12
   - 2.3 Constraints and assumptions ..... 12
3. Stakeholder ..... 13
4. High-level requirements ..... 14
   - 4.1 Context of the system ..... 14
   - 4.2 Functional requirements ..... 14
   - 4.3 Non-functional requirements ..... 15
   - 4.4 Other requirements and constraints ..... 15
5. Solution approach ..... 16
   - 5.1 Architecture overview ..... 16
   - 5.2 How the research shaped the design ..... 17
   - 5.3 Explored alternative solutions (not chosen) ..... 17
   - 5.4 Technology decisions still open ..... 17
   - 5.5 Delivery approach and handover ..... 17
6. Product backlog ..... 18
   - 6.1 Backlog and release plan ..... 18
   - 6.2 Revised projected schedule ..... 19
7. Quality plan ..... 20
   - 7.1 Quality expectations ..... 20
   - 7.2 Quality assurance steps ..... 21
   - 7.3 Planned testing ..... 21
   - 7.4 Accuracy evaluation method ..... 21
   - 7.5 Defects and exit criteria ..... 22
8. Research and investigations ..... 22
   - 8.1 Application domain: enterprise knowledge and RAG ..... 22
   - 8.2 Paper 1 – Chatbot System Architecture (Mohammed and Aref) ..... 22
   - 8.3 Paper 2 – A Repository-First Framework for Democratizing Chatbot Development (Boonsri and Buranasaksee) ..... 23
   - 8.4 Similar systems and alternatives ..... 23
   - 8.5 Technology investigation ..... 23
   - 8.6 Security and identity research ..... 24
   - 8.7 Research responsibilities and open questions ..... 25
9. Skills analysis ..... 25
10. Ethical considerations ..... 26
    - 10.1 Applicable laws, principles and standards ..... 26
    - 10.2 Ethical risks and controls ..... 27
    - 10.3 Data handling rules for the project ..... 28
    - 10.4 Human oversight and transparency ..... 28
    - 10.5 Use of generative AI in preparing this document ..... 28

**References** ..... **29**

**Appendices** ..... **30**
Appendix A. Revised projected schedule (work breakdown) ..... 31
Appendix B. Proposed outline of the Technical Specification Document ..... 34
Appendix C. Requirements-to-test traceability matrix ..... 35
Appendix D. Task board snapshot (Weekly Tasks, Weeks 5–6) ..... 36
Appendix E. Glossary ..... 38

# Acknowledgment of Country

First and foremost, we acknowledge the Traditional Owners of the lands on which Swinburne University of Technology stands, the Wurundjeri Woi Wurrung people of the Kulin Nation, and pay our respects to their Elders past and present.

All members of our team completed this work while living in Vietnam, so none of us live on Aboriginal or Torres Strait Islander land. In addition, we recognise the continuing connection of Aboriginal and Torres Strait Islander peoples to Country, culture and community, and we respect their place in the Australian university where we are studying.

# PART 1 Team Code of Conduct

## 1. Team profile

This section describes the six members of the team and summarises their profiles in Table 1. The team has worked together for six weeks, so each profile is built from three kinds of evidence: (i) the major recorded in the team roster, (ii) the sections and set-up tasks each member has completed on the shared task board in Weeks 5 and 6 (Proposal, Charter, repository, sandboxes and planning documents), and (iii) the working pattern visible in the sprint syncs. Self-assessed proficiency has not yet been collected. Wherever a skill has not been demonstrated, it is described as an area of study or a planned responsibility rather than as expertise, and each member is asked to confirm or correct their row before the Alpha review (see Part 2, Section 9).

- **Truong Ngoc Gia Hieu** (project manager). Created the GitHub repository, the branching rules and the Slack and Confluence sandboxes, and wrote the Proposal summary, Charter scope and risk sections, the Product Backlog and the Risk register. Acts as the primary contact for the sponsor, Mr. Phat (Eric) Pham, and consolidates the team documents.
- **Pham Nguyen Minh Hoang**. Investigates the Slack framework (Bolt SDK, Socket Mode, OAuth scopes) in Task 27 and wrote the Proposal benefits, the Charter business case and resources, and the stakeholder, team-role and skills sections. Proposed owner of the Slack interface.
- **Nguyen Quang Khai**. Wrote the Proposal action plan, the Charter purpose and tentative schedule, and the scope and research sections. Supports the Slack investigation and is the proposed owner of Confluence ingestion.
- **Lu Nhat Hoang** (Software Development major). Wrote the Proposal schedule, the Charter goals and costs, the Week 6 project schedule, the teamwork roadmap, the high-level requirements and the ethics section. Proposed owner of architecture and security, and backup for the Slack framework.
- **Duong Nguyen Dang** (Artificial Intelligence major). Wrote the Proposal budget, the Charter deliverables and benefits, the solution approach and the document-management section, and produced the presentation slides. Proposed owner of retrieval and the demonstration.
- **Le Xuan Ny** (Artificial Intelligence major). Wrote the Proposal desired outcome, the Charter problem and out-of-scope sections, the meeting minutes and the Confluence upload of the governance documents. Proposed owner of quality assurance and bilingual validation.

**Table 1: Team Profile**

| Student name | Technical skills | Soft skills | Communication | Teamwork | Other |
|---|---|---|---|---|---|
| **Truong Ngoc Gia Hieu (105565520)** | Git/GitHub workflow and branch protection; Slack app (Bolt, Socket Mode) and Confluence sandbox configuration at set-up level. Major not recorded. Python proficiency to be confirmed. | Project planning, risk management, client liaison, consolidating documents. | Main contact with Mr. Phat; sends written questions in advance; coordinates on Slack, Discord and Google Meet. | Initiator and coordinator; also edits sections written by others. | Maintains the shared Excel board and Google Workspace documents. |
| **Pham Nguyen Minh Hoang (105543500)** | Slack Bolt SDK, Socket Mode, OAuth scopes and Python event handlers (under investigation in Task 27). Major not recorded. | Analytical writing (business case, stakeholder analysis, skills analysis). | Written reports and shared documents; to be confirmed. | Owner of the Slack workstream; works with Khai and Nhat Hoang. | Learns from official documentation and video tutorials. |
| **Nguyen Quang Khai (105560512)** | Requirements and architecture research; Confluence ingestion is a planned responsibility. Major not recorded. Parsing skills to be demonstrated. | Structured documentation, scheduling and scoping. | Written; to be confirmed. | Research and documentation contributor; supports the Slack investigation. | Proposed ingestion and technical research lead. |

| Student name | Technical skills | Soft skills | Communication | Teamwork | Other |
|---|---|---|---|---|---|
| **Lu Nhat Hoang (105234956)** | Software Development major: software design, interfaces and integration. Security and SQL skills to be demonstrated. | Scheduling, requirements and ethics analysis. | Written specifications; to be confirmed. | Planner and architecture owner; backup for the Slack framework. | Proposed software architecture and security coordinator. |
| **Duong Nguyen Dang (105508444)** | Artificial Intelligence major: embeddings, retrieval and grounded answers (framework proficiency unverified). | Presentation design (Canva slides) and process documentation. | Presents project progress to the group and supervisor. | Contributor with a demonstration role. | Proposed RAG implementation and demonstration lead. |
| **Le Xuan Ny (105680151)** | Artificial Intelligence major; Confluence page management (minutes and document upload). | Minute-taking, quality checking, tracking actions. | Records decisions in the Confluence minutes. | Recorder and reviewer; backup for retrieval and ingestion. | Proposed quality assurance and validation lead. |

*Sources: team roster (majors) and the task board, Weeks 5 and 6. Entries marked "to be confirmed" are open on purpose and are not assumptions about the member.*

## 2. Team role

Table 2 proposes a primary role and backup contribution for each member using the current task allocation and recorded majors. Gia Hieu's project-manager role is established in the Project Charter. The remaining role labels are recommendations for member confirmation when Table 1 is completed; task ownership demonstrates responsibility, not verified technical proficiency.

**Table 2: Team Role**

| Student name | Team role | Justification |
|---|---|---|
| **Truong Ngoc Gia Hieu** | Project manager and integration coordinator | The Charter identifies him as project manager. His repository, backlog and client-coordination tasks support planning, review and delivery accountability. He coordinates decisions across workstreams. |
| **Pham Nguyen Minh Hoang** | Slack interface and integration lead (proposed) | Task 27 assigns him the Slack framework investigation, supported by Khai and Nhat Hoang. He owns the mention-response prototype, thread behavior, integration guide and live connectivity demonstration. |
| **Nguyen Quang Khai** | Ingestion and technical research lead (proposed) | His action-plan, scope, team-profile and research tasks provide context for source requirements and architecture investigation. He can lead the Confluence parsing spike and support Minh Hoang; implementation proficiency needs demonstration. |
| **Lu Nhat Hoang** | Software architecture and security coordinator (proposed) | His recorded Software Development major and schedule, high-level requirements and ethics assignments support interface design, integration planning and access-control review. He backs up the Slack framework. |
| **Duong Nguyen Dang** | RAG implementation and demonstration lead (proposed) | His recorded AI major and solution-approach/presentation tasks align with investigating embeddings, retrieval and grounded answers. He can lead the retrieval spike and demonstrate results; specific framework expertise remains unverified. |
| **Le Xuan Ny** | Quality assurance and knowledge validation lead (proposed) | Her recorded AI major and problem-statement, meeting-minutes and quality-plan tasks support test design, requirement traceability and source validation. She backs up retrieval/ingestion and coordinates bilingual evaluation. |

## 3. Teamwork roadmap

### 3.1 Weekly work pattern and communication protocol

- Hold formal sprint syncs every Monday and Thursday at 20:00 via Google Meet or Microsoft Teams.
- Hold a fortnightly check-in with the sponsor, Mr. Phat, on Thursdays (29 Oct, 12 Nov and 26 Nov 2026); send written questions in advance and record decisions in the meeting minutes.

- Use Slack and Discord for informal coordination and Google Workspace for collaborative documentation.
- Expect urgent responses on Slack or Discord within 2 to 4 hours, and 24 hours for general updates.
- Complete and upload internal task deliverables 48 hours before official university submission windows.
- Conduct peer code reviews and two-pass documentation checks for all revisions.

### 3.2 Task estimation, assignment and monitoring

- Break tasks down into the four architectural layers (presentation, security, RAG core, ingestion and storage) plus the quality and documentation workstreams, and estimate them from technical complexity and workflow dependencies.
- Distribute tasks according to individual engineering expertise and development goals; every task has a primary and a backup owner.
- Track work items, deadlines and accountability in real time on the shared project management board. The board is the Excel workbook "Computing Technology Project A.xlsx": the sheet Weekly Tasks holds each task with its objective, owner, dates, status and materials, and the sheet Project_Scheduled holds the WBS, the Gantt chart and the revised schedule. A snapshot is given in Appendix D. The team chose a shared workbook because it is free, visible to everyone and sits next to the Google Workspace documents.

### 3.3 Communication rules, expected behaviour and fallback plans

- Practice proactive transparency by reporting roadblocks immediately rather than waiting for scheduled meetings.
- Maintain mutual accountability, punctuality and respect during code reviews and architectural critiques.
- Re-allocate tasks at the next sprint sync, or immediately on Slack or Discord when the matter is urgent, if a member faces illness or unexpected absence.
- Fall back to local execution paths (such as local Socket Mode and Ollama) if cloud free-tier services experience downtime.

## 4. Document management

Shared documents, codebase and project artifacts are managed and version-controlled using a structured, multi-platform approach to ensure collaboration efficiency and data integrity.

- **Code and technical artifacts:** All source code, configuration scripts and technical pipelines are managed in the official GitHub repository. The team enforces a feature-branching workflow (feature/*) with pull requests and at least one mandatory peer review before merging into the main development branch (develop). Direct pushes to main are not allowed; main is reserved for showcase-ready releases. Semantic commit messages are used to keep a clear change log and reduce merge conflicts.
- **Project documentation and reports:** Core project documents (the Project Charter, Project Proposal, quality plans and technical notes) are co-authored and stored in shared Google Workspace tools (Docs, Sheets and Slides), which give real-time collaboration, built-in revision tracking and central access.

- **Meeting minutes and decisions:** Client interactions, feedback from the project sponsor Mr. Phat (Eric) Pham, weekly sprint reviews and architectural decisions are documented in shared meeting-minute logs on Confluence and linked to the task board for traceability.
- **Conventions:** Each document section has one owner and one reviewer recorded on the task board; Gia Hieu consolidates the final document. Files are named COS40005_<Document>_v<major>.<minor>, and a document is only submitted after the two-pass check (content, then format and references).

## 5. Risk mitigation

Effective risk management is essential to delivering this project within a fixed twelve-week timeline and a zero-cost constraint. The team identified risks from skill gaps, team availability and project-specific issues such as delayed Confluence access, LLM latency and data leakage. Each risk is recorded in Table 3 with its impact, mitigation plan and owner. The register is reviewed at the start of each weekly team meeting and updated at the end of every sprint. Likelihood (L) and Impact (I) are rated Low (L), Medium (M) or High (H). The owners named "Ingestion lead", "RAG lead" and "Security lead" are Khai, Dang and Nhat Hoang respectively (Table 2). Risks R16 and R17 were added when the schedule was revised on 25 October 2026.

**Table 3: Risk register**

| ID | Risk | L | I | Impact on team/project | Mitigation plan | Owner |
|---|---|---|---|---|---|---|
| R1 | Skill gap in RAG/LLM tooling (LangChain/LlamaIndex, embeddings, reranking) | H | H | Slow development, poor retrieval accuracy (target ≥80%) | Hold a 1–2 week learning spike in Sprint 1. Pair a stronger member with a less experienced one. Build the pipeline from small prototypes first. Record findings in the shared wiki. | Gia Hieu |
| R2 | Skill gap in Slack Bolt, Socket Mode and SSO | M | M | Delayed bot prototype and security features | Follow official Slack docs and build a minimal echo-bot in Sprint 1. Assign one member as Slack "specialist" and have them share a short demo. | All |
| R3 | Team member unavailable (illness, exams, other units, travel) | M | H | Tasks blocked, workload falls on others | Every task has a primary and a backup owner. Keep documentation current so anyone can pick up work. Members must notify the team 24h ahead of planned absence. Re-assign tasks at the next meeting. | All |
| R4 | Uneven contribution / poor communication | M | M | Missed deadlines, team conflict | Use a visible task board with weekly check-ins. Follow the response-time rule (reply within 24h). Escalate to the unit coordinator or tutor if unresolved after two warnings. | Gia Hieu |
| R5 | Confluence access delayed or restricted (permissions, API tokens, VIB policy) | M | H | Cannot build or test ingestion pipeline | Request access early via Mr. Phat. Prepare a sample dataset of exported Confluence pages (public or low-level content) as a fallback. Design the ingestion layer so the source is swappable. | Ingestion lead |
| R6 | Messy Confluence content (HTML tables, macros, embedded images) | H | M | Bad Markdown, poor chunks, wrong answers | Build standardised REST parsing and HTML→Markdown converters. Test on a variety of page types. Log pages that fail parsing. Handle images in a separate step. | Ingestion lead |
| R7 | LLM API throttling, latency or free-tier limits | M | H | Response time exceeds the 3-second target, demo failures | Cache frequent responses and optimise retrieval (top-k tuning). Keep a local open-source model as a fallback. Measure latency from Sprint 3 onward. | RAG lead |
| R8 | Insufficient local hardware for local LLM/embeddings | M | M | Slow indexing, crashes | Use smaller or quantised models. Share one indexed vector DB snapshot between members. Containerise the environment for consistency. | RAG lead |
| R9 | Weak Vietnamese/English retrieval quality | M | H | Accuracy below 80% target for one language | Use a multilingual embedding model. Build a bilingual test query set. Tune chunk size and prompts per language. | RAG lead |

| ID | Risk | L | I | Impact on team/project | Mitigation plan | Owner |
|---|---|---|---|---|---|---|
| R10 | Data leakage or channel whitelist bypass | L | H | Breach of the 100% whitelist compliance target, loss of client trust | Implement the whitelist early (Sprint 2). Deny by default. Write negative test cases (non-whitelisted channels and users). Never commit tokens, so use .env and secret scanning. Use only non-sensitive documents for development. | Security lead |
| R11 | Hallucination or answers without citations | M | H | Incorrect information presented as fact | Use strict system prompts that require source citations. Add a fallback reply when no relevant context is found. Review answers against a test set. | RAG lead |
| R12 | Scope creep (extra data sources, production deployment) | M | M | Schedule slippage | Keep the Charter's "outside of scope" list as the baseline. Any new item goes through the Product Backlog and is approved by the client before it enters a sprint. | Gia Hieu |
| R13 | Schedule slippage / tight timeline | H | H | Incomplete Alpha release | Plan work in 2-week sprints. Prioritise High-value items first. Review velocity at each sprint review and cut Low-priority items if behind. Sprint 3 (Weeks 9–10) is the heaviest sprint: B12 and B14 are the first items to cut. | Gia Hieu |
| R14 | Merge conflicts / lost work | M | M | Rework, broken builds | Use GitHub with feature branches and pull requests (at least one review). Follow a modular folder structure. Merge small and often. | All |
| R15 | Client feedback delays | M | M | Requirements unclear, rework | Book a fixed fortnightly check-in with Mr. Phat. Send written questions in advance. Record decisions in meeting minutes. | Gia Hieu |
| R16 | Technical Specification incomplete or unclear for VIB's deployment team | M | H | Sponsor cannot deploy from the document; rework in the final week | Agree the specification outline with Mr. Phat on 12 Nov. Draft sections in parallel with the build from Week 9. Walk the sponsor through the draft on 26 Nov. Send v1.0 on 2 Dec, leaving four days for questions. Apply the two-pass review. | Gia Hieu |
| R17 | Deployment depends on VIB approvals and environment (Slack admin, identity provider, hosting) | M | M | Spec assumes facilities that VIB does not provide | List prerequisites and assumptions explicitly in the specification. Confirm them at sponsor check-ins. Test the documented deployment steps on the sandbox. Keep Socket Mode as the documented baseline and describe the HTTP Events alternative. | Gia Hieu / Minh Hoang |

# PART 2 Project overview and plan

## 1. Problem statement

### 1.1 Background

Vietnam International Bank (VIB) keeps its internal policies, procedures and technical documentation in Confluence. Staff must search several spaces by keyword, open long pages and read through tables, macros and attachments to find a single answer. When the search fails, they ask colleagues in Slack, which interrupts other people and produces answers that cannot be traced back to an approved page. In a bank, the answer must also come only from content that the asker is allowed to see, so any assistant has to respect channel and identity boundaries from the first release.

### 1.2 Problem

Internal knowledge is fragmented across Confluence spaces and is hard to retrieve through keyword search. The result is slow look-ups, repeated questions to colleagues and a risk that employees rely on outdated or unverified answers. Existing commercial enterprise-search products (for example Slack AI and Atlassian Rovo) were considered but are outside the zero-cost constraint of this project and give limited control over how banking documents are processed. A general-purpose chatbot without retrieval cannot answer from VIB's own, frequently updated pages and tends to invent answers.

### 1.3 Aim, purpose and objectives

**Aim.** To design and build a proof-of-concept Slack assistant that answers English and Vietnamese questions from Confluence content, always showing its sources, and that only works in approved channels and for validated users; and to hand over a Technical Specification Document that lets VIB deploy the system.

**Purpose.** To turn static documentation into an active, conversational knowledge service that shortens search time, reduces context switching and keeps access governed, while testing at no extra cost whether the approach is feasible for an enterprise rollout.

The objectives in Table 4 are measurable. The targets are taken from the Project Charter and are validation targets, not achievements: they will be measured in Week 10 and reported in the Technical Specification Document.

**Table 4: Project objectives and success measures**

| ID | Objective | Measure of success | Target | Planned |
|---|---|---|---|---|
| O1 | Ingest approved Confluence pages (prose, tables, images, restricted pages) into a searchable index. | Pages of the agreed mock set processed without manual repair; parse failures logged. | All agreed pages indexed | Weeks 7–8 |
| O2 | Answer questions in English and Vietnamese from the indexed content. | Answer accuracy on the agreed bilingual test set (definition in Part 2, Section 7). | ≥80% in each language | Week 10 |
| O3 | Respond quickly enough for conversation in Slack. | Average end-to-end response time on stated hardware (median and 95th percentile also reported). | <3 seconds average | Week 10 |
| O4 | Answer only inside approved channels and for validated users. | Negative tests with non-whitelisted channels, users and restricted documents. | 100% compliance | Week 10 |
| O5 | Make every answer verifiable. | Share of answers with at least one correct source link, or a fallback reply. | 100% of answers | Weeks 9–10 |
| O6 | Hand over a specification VIB can deploy from. | Technical Specification Document sent to Mr. Phat; his questions answered and revisions agreed. | Sent 2 Dec; final 6 Dec 2026 | Weeks 11–12 |
| O7 | Stay inside the project constraints. | Additional spend; data used for development. | Zero additional spend; mock or approved data only | Whole project |

# 2. Scope

## 2.1 In scope

- A Confluence ingestion pipeline using the REST API for the sandbox space VIB-Mock-Knowledge and any pages later approved by VIB: page fetching, HTML-to-Markdown conversion (tables and macros), basic handling of embedded images, chunking, multilingual (English/Vietnamese) embedding and indexing in a local vector database.
- A RAG engine with a retriever, an LLM (local open-source or an approved API), prompt templates that require source citations, English/Vietnamese query handling and a fallback reply for unanswerable questions.
- A Slack app built with the Bolt SDK and Socket Mode that answers @mentions in threads, formats answers with Block Kit and shows clickable source links.
- Security middleware: channel whitelist (deny by default) and Slack identity (SSO) validation before a query reaches the RAG engine.
- Testing and benchmarks: bilingual accuracy test set, latency benchmark with response caching, and negative security tests.
- Documentation and handover: architecture documents, test report, demo script, handover pack and the **Technical Specification Document** sent to Mr. Phat in Weeks 11–12.

## 2.2 Out of scope

- Channels or platforms other than Slack (for example Microsoft Teams, email or a web chat) and sources other than Confluence in Project A (GitHub repositories and internal documents are Beta items).
- Beta features: context reranking, conversation memory, full role-based access control, incremental synchronisation, audit logging and user feedback buttons (backlog items B19–B25).
- Integration with core banking systems, transactions or any automated workflow, and multi-tenant role management.
- Production deployment, hosting and support in VIB's environment. VIB deploys from the Technical Specification Document; any deployment by the team needs a separately approved scope (assumption to confirm with Mr. Phat).
- Paid cloud infrastructure, paid LLM quotas and continuous real-time synchronisation of the knowledge base.
- Processing of real VIB production or confidential data before written approval from the Confluence administrators, Slack administrators and VIB IT/security.
- Fine-tuning or training new language models.

## 2.3 Constraints and assumptions

- Zero additional spend: free tiers, open-source tools and student hardware only. Free services and Socket Mode do not guarantee continuous hosting.
- A fixed twelve-week timeline; Project A (Alpha release) covers Weeks 5–12.
- Development and testing use the Slack and Confluence sandboxes and mock or low-sensitivity content.
- Mr. Phat confirms permitted test data, the content of the Technical Specification Document and the acceptance criteria at the sponsor check-ins.

## 3. Stakeholder

The project has the following stakeholders. Named governance roles follow the Project Charter; operational roles without named representatives must be nominated by the client before enterprise testing.

- **Mr. Phat (Eric) Pham** — project sponsor, client representative and process owner. He clarifies the business problem, confirms permitted test data and scope, reviews demonstrations, receives the Technical Specification Document, and accepts the PoC and handover. Gia Hieu consolidates questions and records his decisions in meeting minutes.
- **Truong Ngoc Gia Hieu** — project manager and primary client liaison. He coordinates the six-member team, maintains priorities and risks, manages repository workflow, and escalates scope or schedule changes to the sponsor.
- **Development team** — Duong Nguyen Dang, Nguyen Quang Khai, Lu Nhat Hoang, Le Xuan Ny, Pham Nguyen Minh Hoang and Truong Ngoc Gia Hieu. The team delivers the Slack interface, ingestion, retrieval, security controls, tests and documentation. Proposed workstream ownership is recorded in Table 2 and must be reviewed as implementation skills become demonstrable.
- **VIB employees and department staff** — intended end users. They need relevant Vietnamese/English answers, traceable source links and safe access within their authorized context. Representative users should review realistic mock queries and report incorrect, missing or unclear answers during pilot evaluation.
- **Knowledge owners and Confluence administrators** — maintain source accuracy, classify documents, approve ingestion access and clarify page restrictions. They review representative pages, tables and image content, and confirm whether source updates or access changes require reindexing. Named representatives are pending client nomination.
- **Slack workspace administrators and VIB IT/security representatives** — approve app installation, scopes, permitted channels and identity/access design. They review negative authorization tests, secrets handling and deployment prerequisites. Sandbox access does not imply approval to process production banking information.
- **Course lecturer, tutor and academic supervisor** — assess the Team and Project Plan, research, contribution evidence and Alpha deliverable against COS40005 requirements. The team's named academic contact is not confirmed in the available records; Gia Hieu should record it with the review schedule.
- **Future system maintainer or operations owner** — receives the Technical Specification Document, configuration instructions, source code, test evidence and known limitations. This role contributes to handover readiness; production hosting and ongoing support remain subject to a separately approved scope.

**Engagement and acceptance:** use the existing task board, shared Google Docs and meeting minutes for review evidence. The sponsor approves business scope and acceptance; administrators approve access; academic staff assess coursework. API vendors and model providers are external dependencies, not project acceptance authorities. Current Charter targets (at least 80% retrieval/answer accuracy, average response below 3 seconds, and 100% whitelist compliance) are validation targets rather than measured achievements.

# 4. High-level requirements

## 4.1 Context of the system

The system is a **new prototype** (proof of concept): it does not upgrade or replace an existing product, because VIB staff currently find answers by searching Confluence and asking colleagues. It is **a component of a larger environment** rather than a standalone product. It sits between three existing platforms: Slack as the user interface, Confluence as the source of knowledge, and a language model that writes the answer from retrieved text. Figure 1 shows how these parts relate. The Alpha release (Project A) delivers the proof of concept described in this plan; the Beta release (Project B) extends it with reranking, role-based access control and further sources.

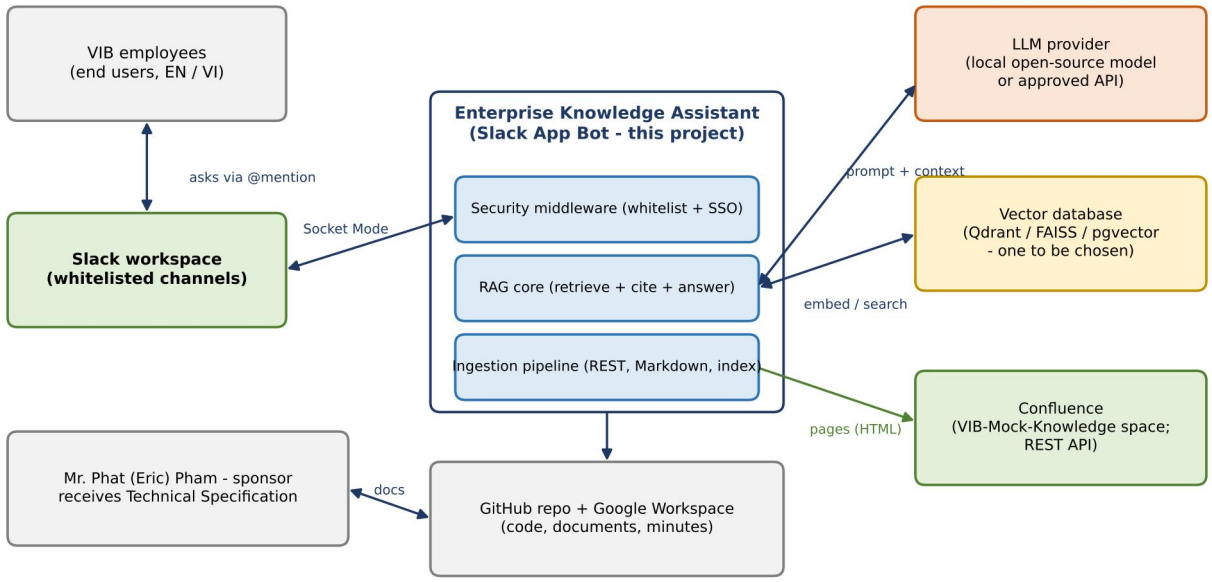



**Figure 1: System context of the Slack-based knowledge assistant**

## 4.2 Functional requirements

**Table 5: Functional requirements (Alpha release)**

| ID | Requirement | Value | Backlog | Status |
|---|---|---|---|---|
| FR1 | Fetch pages from configured Confluence spaces through the REST API, keeping title, URL, last-modified date and page-restriction metadata. | H | B1 | Planned W7 |
| FR2 | Convert page HTML, including tables and macros, to clean Markdown and log pages that fail to parse. | H | B3 | Planned W8 |
| FR3 | Handle embedded images (associate image assets with pages; OCR scope to be confirmed with the sponsor). | M | B12 | Planned W9 |
| FR4 | Split text into chunks, create multilingual (English/Vietnamese) embeddings and index them in the vector database. | H | B2, B4 | Planned W7–8 |
| FR5 | Respond to @mentions in Slack in a thread, through a Bolt app running in Socket Mode. | H | B5, B10 | Planned W7, W9 |
| FR6 | Retrieve the most relevant chunks and generate an answer from them with an LLM. | H | B7 | Planned W9 |
| FR7 | Include at least one clickable source citation to the Confluence page in every answer. | H | B8, B14 | Planned W9–10 |
| FR8 | Detect the language of the question and answer in the same language. | H | B9 | Planned W9 |
| FR9 | Reply with a clear fallback message when no relevant context is found. | M | B13 | Planned W10 |
| FR10 | Process requests only from whitelisted channels; deny every other channel by default. | H | B6 | Planned W7 |

| ID | Requirement | Value | Backlog | Status |
|---|---|---|---|---|
| FR11 | Validate the identity of the asking user (Slack SSO) before answering. | H | B11 | Planned W8 |
| FR12 | Format answers with Slack Block Kit for readability. | M | B14 | Planned W10 |
| FR13 | Cache frequent responses to protect response time. | H | B16 | Planned W10 |
| FR14 | Keep whitelist, spaces, model and index settings in configuration files or environment variables, not in code. | H | B5, B6 | Planned W7 |

*Beta requirements (not in Alpha): context reranking (B19), conversation memory (B20), role-based access control (B21), incremental sync (B22), new data sources (B23), audit logging (B24), feedback buttons (B25) and production-readiness plan (B26).*

## 4.3 Non-functional requirements

### Table 6: Non-functional requirements

| ID | Quality | Requirement | How it is verified |
|---|---|---|---|
| NFR1 | Accuracy | At least 80% accuracy on the agreed bilingual test set, in each language. | Accuracy benchmark (B15), two independent graders |
| NFR2 | Performance | Average response time below 3 seconds, measured end to end on stated hardware. | Latency benchmark (B16): mean, median, 95th percentile |
| NFR3 | Security | 100% compliance with the channel whitelist; no tokens or secrets in the repository. | Negative tests (B17); secret scanning on every pull request |
| NFR4 | Privacy | Only mock or approved low-sensitivity data in development; no personal data in logs or repository. | Data-handling checklist at each sprint review |
| NFR5 | Language | English and Vietnamese handled with comparable quality. | Per-language results in the benchmark |
| NFR6 | Reliability | Graceful behaviour when the LLM or vector database is unavailable; Socket Mode reconnects after a drop. | Failure-injection tests; reconnection evidence |
| NFR7 | Portability and modularity | Swappable source, vector store and LLM adapters; environment containerised. | Interface contracts and mock-backed tests (Week 8) |
| NFR8 | Maintainability | Readable code, reviewed pull requests, tests for parsers and middleware, documented configuration. | Peer review; Technical Specification checklist |
| NFR9 | Cost | Zero additional spend. | Cost log at each sprint review |
| NFR10 | Usability | Short, readable answers with visible sources in Slack. | Review of realistic queries with representative users |
| NFR11 | Traceability | Each requirement is linked to a backlog item and at least one test. | Traceability matrix (Appendix C) |

## 4.4 Other requirements and constraints

- **Deliverables:** Alpha code and tests, benchmark results, demo script, handover pack and the Technical Specification Document (Appendix B gives its proposed outline).
- **Academic:** the plan, research, contribution evidence and Alpha release are assessed against COS40005 requirements.
- **Time:** the Technical Specification Document is sent to Mr. Phat on 2 December 2026 and the project is handed over on 6 December 2026.
- **Governance:** any new item goes through the Product Backlog and needs the sponsor's approval before it enters a sprint.

# 5. Solution approach

## 5.1 Architecture overview

The proposed system architecture implements a secure, modular Retrieval-Augmented Generation (RAG) pipeline integrated natively into a corporate Slack workspace. As illustrated in Figure 2, the solution is structured into four core layers.

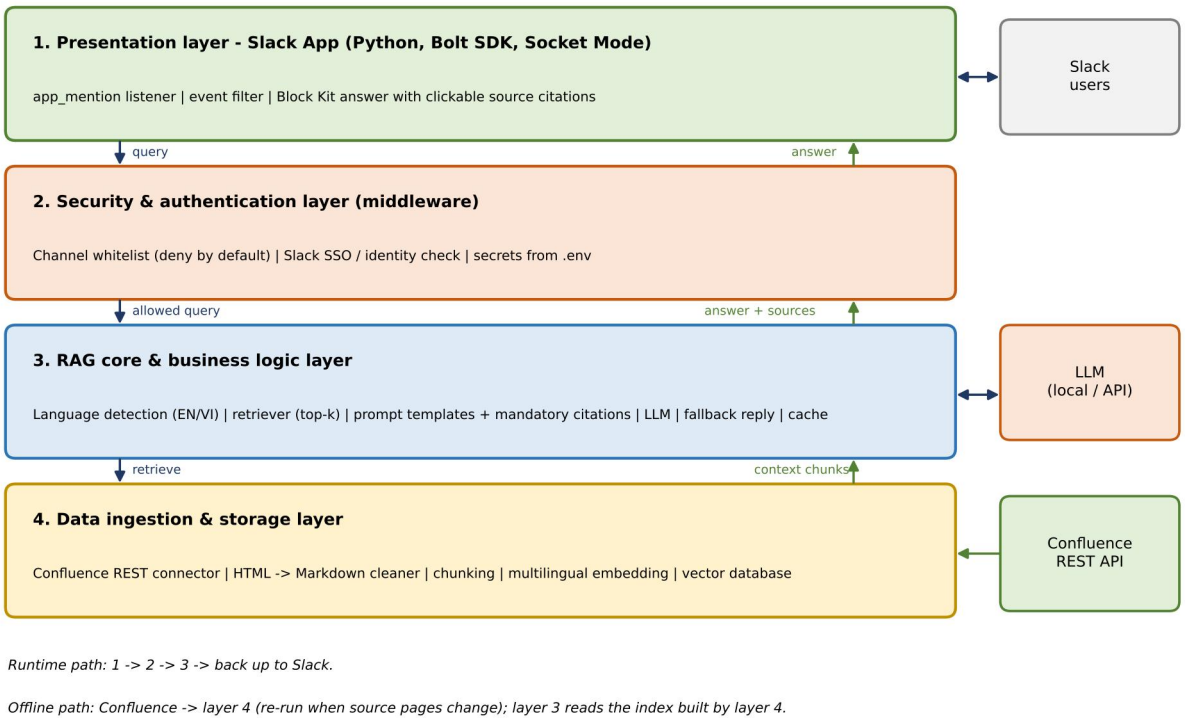



**Figure 2: Four-layer solution architecture and data flow**

- **Presentation Layer (Slack App Interface):** Built using Python, the Slack Bolt SDK and Socket Mode. Socket Mode establishes a secure, persistent WebSocket connection, eliminating the need for public webhooks or live server deployments during the Proof of Concept (PoC) phase. It listens for user mentions (app_mention), filters events, and delivers formatted responses via Slack Block Kit with clickable source citations.
- **Security & Authentication Layer (Middleware):** Enforces strict access governance through channel-level whitelisting (deny-by-default) and Slack Single Sign-On (SSO) validation. Incoming queries are filtered to ensure the bot operates exclusively within authorized enterprise channels and authenticated user contexts.
- **RAG Core & Business Logic Layer:** Processes sanitized queries through a retrieval engine connected to a vector database and a Large Language Model (for example AWS Bedrock or local open-source models). Custom prompt templates enforce mandatory source citations, manage bilingual (English/Vietnamese) query processing, and provide fallback responses for unanswerable inquiries.
- **Data Ingestion & Storage Layer:** Features an automated REST API pipeline that extracts documentation pages from Confluence workspaces. Raw HTML content passes through a cleaning converter (handling tables, macros and embedded images) before being chunked, embedded using multilingual models, and indexed into a vector database (for example Qdrant, Chroma or FAISS).

## 5.2 How the research shaped the design

Two papers were studied in detail (Part 2, Section 8). The design borrows the following from them:

- **Strict layering with swappable adapters.** The repository-first framework of Boonsri and Buranasaksee [2] uses four layers with inward dependencies and a small adapter interface so that channels, data sources and LLM providers can be replaced without touching the core. We apply the same idea: the Confluence source, the vector store and the LLM are accessed through adapters, which also answers risk R5.
- **Pipeline of understanding, decision and generation.** The generic chatbot architecture of Mohammed and Aref [1] separates natural language understanding, a dialogue manager and a language generator. Our language detection and query clean-up play the NLU part, the retrieval strategy with a fallback reply is the dialogue manager, and the cited LLM answer with a content check is the generator (mapping in Part 2, Section 8.2).
- **Acceptance checklists.** The same framework paper grades results as pass, partial or fail against a written checklist; we reuse this for the accuracy benchmark (Part 2, Section 7).

## 5.3 Explored alternative solutions (not chosen)

- **Commercial enterprise search (Slack AI / Atlassian Rovo):** evaluated as out-of-the-box alternatives, but discarded because of high subscription costs (Business+/Enterprise+ tiers) and insufficient customisation flexibility for internal banking document structures under the zero-cost PoC constraint.
- **HTTP Events API (webhook-based architecture):** considered for Slack communication, but not chosen for the local development PoC because it requires public IP addresses, domain registration and SSL configuration. Socket Mode was selected instead to enable secure, real-time local development. The specification will describe the HTTP option for production hosting.
- **Direct LLM prompting without retrieval:** considered for quick implementation, but rejected because of high hallucination rates and the inability to query proprietary, frequently updated internal Confluence documentation, which would fail the ≥80% accuracy target.

## 5.4 Technology decisions still open

Two choices are not yet final and will be recorded in a decision record in Week 7: (a) the vector store, where Qdrant or FAISS (earlier choice) compete with PostgreSQL with pgvector (newer proposal), and (b) the LLM, local open-source via Ollama or an approved API. The decision rule is: the smallest stack that meets the accuracy and <3-second targets on the team's hardware at zero cost, with metadata filtering for access control.

## 5.5 Delivery approach and handover

Project A is delivered in three steps, shown in Figure 3 and Section 6.2. **Delivery Phase 1** (Weeks 7–8) builds the data foundation, the Slack gateway and the security middleware. **Delivery Phase 2** (Weeks 9–10) builds the RAG core, connects it to Slack and validates accuracy, latency and security. The remaining weeks (11–12) are used to write the **Technical Specification Document**, which is sent to Mr. Phat on 2 December so that VIB can deploy the solution from it. The specification is drafted section by section while components are built, so that Weeks 11–12 are for consolidation, review and the sponsor's questions.

# 6. Product backlog

## 6.1 Backlog and release plan

The Product Backlog translates the project's high-level requirements into a prioritised list of deliverables. Items are grouped into epics and given a business value (H = must-have for the PoC, M = important, L = nice-to-have) together with their dependencies. Each item is allocated to a sprint; Sprints 1 to 4 form Project A (Weeks 5–12) and Sprints 5 to 8 form Project B. Each sprint lasts two weeks. Sprint 1 (Weeks 5–6) was used for initiation, the repository, the sandboxes and the architecture decisions, so build work starts in Sprint 2 (Week 7).

The Alpha release delivers a functional Proof of Concept that answers English and Vietnamese queries from Confluence through Slack with channel whitelisting and SSO validation; Sprint 4 turns the results into the Technical Specification Document for Mr. Phat. The Beta release adds advanced retrieval, role-based access control and additional data sources. The backlog is refined at each sprint review in consultation with the client. Compared with the first version of this backlog, the sprint allocation was revised on 25 October 2026 to match the new schedule: items B1 and B2 moved from Sprint 1 to Sprint 2, B11 moved from Sprint 3 to Sprint 2, B13 to B17 moved from Sprint 4 to Sprint 3, and B18 now centres on the Technical Specification Document.

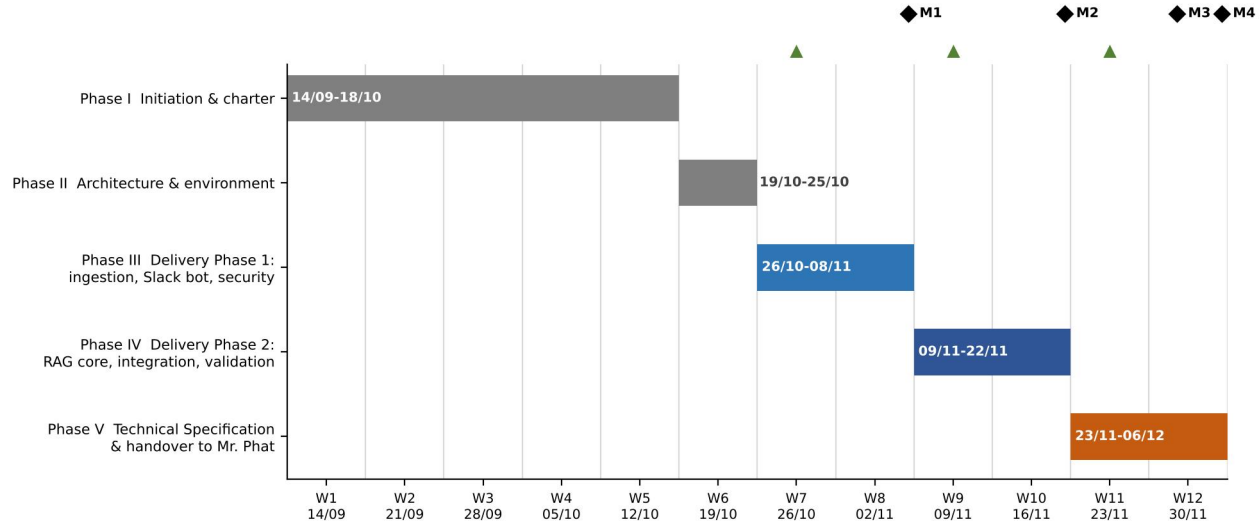
**Table 7: Product backlog**

| ID | Epic | Backlog item | Value | Depends on | Sprint | Release |
|---|---|---|---|---|---|---|
| B1 | Ingestion | Confluence REST API connector to fetch pages from selected spaces | H | None | S2 · W7 | Alpha |
| B2 | Storage | Set up the local vector database and containerised dev environment | H | None | S2 · W7 | Alpha |
| B3 | Ingestion | HTML→clean Markdown converter (tables, macros) | H | B1 | S2 · W8 | Alpha |
| B4 | Ingestion | Chunking and multilingual embedding (EN/VI) and indexing | H | B2, B3 | S2 · W8 | Alpha |
| B5 | Slack | Slack app setup with Bolt and Socket Mode | H | None | S2 · W7 | Alpha |
| B6 | Security | Channel whitelist configuration (deny by default) | H | B5 | S2 · W7 | Alpha |
| B7 | RAG | Retriever and LLM integration (basic Q&A) | H | B4 | S3 · W9 | Alpha |
| B8 | RAG | Prompt templates with mandatory source citations | H | B7 | S3 · W9 | Alpha |
| B9 | RAG | English/Vietnamese query handling | H | B7 | S3 · W9 | Alpha |
| B10 | Slack | Message event handling linked to RAG backend | H | B5, B7 | S3 · W9 | Alpha |
| B11 | Security | Slack SSO user validation | H | B5 | S2 · W8 | Alpha |
| B12 | Ingestion | Embedded image asset handling | M | B3 | S3 · W9 | Alpha |
| B13 | RAG | Fallback response for unanswerable queries | M | B7 | S3 · W10 | Alpha |
| B14 | Slack | Block Kit formatting with clickable source links | M | B10 | S3 · W10 | Alpha |
| B15 | Quality | Test query set and accuracy benchmark (≥80%) | H | B7 | S3 · W10 | Alpha |
| B16 | Quality | Latency benchmark (<3s) and response caching | H | B10 | S3 · W10 | Alpha |
| B17 | Quality | Security boundary tests (100% whitelist compliance) | H | B6, B11 | S3 · W10 | Alpha |
| B18 | Docs | Technical Specification Document sent to Mr. Phat; architecture docs, deployment guide, test report, demo script, handover pack | H | B1–B17 | S4 · W11–12 | Alpha |
| B19 | RAG | Context reranking to improve precision | M | B7 | S5 | Beta |
| B20 | RAG | Conversation memory for follow-up questions | M | B10 | S5 | Beta |
| B21 | Security | Role-based access control (RBAC) | M | B11 | S6 | Beta |
| B22 | Ingestion | Incremental sync and re-indexing of updated pages | M | B1, B4 | S6 | Beta |
| B23 | Ingestion | Add new sources (GitHub repos, internal docs) | M | B22 | S7 | Beta |
| B24 | Security | Audit logging of queries and access | L | B6, B11 | S7 | Beta |
| B25 | Slack | User feedback buttons (helpful / not helpful) | L | B14 | S7 | Beta |

| ID | Epic | Backlog item | Value | Depends on | Sprint | Release |
|---|---|---|---|---|---|---|
| B26 | Deployment | Production-readiness plan and cloud deployment (subject to client approval) | L | B18 | S8 | Beta |

## 6.2 Revised projected schedule

The schedule below replaces the earlier plan in which testing and handover filled Weeks 10–12. Phases I and II (Weeks 1–6) are complete or closing. Phases III and IV give a detailed build plan from Week 7 to Week 10, and Phase V (Weeks 11–12) is the Technical Specification Document and handover to Mr. Phat. Table 8 lists the phases and Figure 3 shows them with the milestones; the full work breakdown with owners is in Appendix A and in the workbook "Computing Technology Project A.xlsx" (sheet Project_Scheduled).




*Diamonds: M1 Phase 1 exit (08/11), M2 Alpha code freeze (22/11), M3 specification sent to Mr. Phat (02/12), M4 final handover (06/12). Triangles: sponsor check-ins (29/10, 12/11, 26/11).*

**Figure 3: Revised projected schedule (Weeks 1–12)**

### Table 8: Phases, milestones and exit criteria

| Phase | Weeks | Dates (2026) | Backlog items | Exit criterion / milestone |
|---|---|---|---|---|
| **I Initiation, Proposal and Charter** | 1–5 | 14/09–18/10 | — | Proposal and Charter signed off by Mr. Phat (complete). |
| **II Architecture design and environment set-up** | 6 | 19/10–25/10 | — | Repository, Slack and Confluence sandboxes, layered architecture, this plan (closing). |
| **III Delivery Phase 1: data foundation, Slack gateway, security** | 7–8 | 26/10–08/11 | B1–B6, B11 | **M1 (08/11):** mock pages ingested and indexed in EN/VI; bot answers @mentions only in whitelisted channels for validated users. |
| **IV Delivery Phase 2: RAG core, Slack integration, validation** | 9–10 | 09/11–22/11 | B7–B10, B12–B17 | **M2 (22/11):** Alpha code freeze; accuracy, latency and security results recorded. |
| **V Technical Specification Document and handover** | 11–12 | 23/11–06/12 | B18 | **M3 (02/12):** specification sent to Mr. Phat. **M4 (06/12):** final demo, revisions and handover. |

*Sponsor check-ins: 29/10 (confirm test data and specification expectations), 12/11 (Phase 1 demo and specification outline), 26/11 (specification walkthrough). VIB deployment from the specification follows handover and is outside the team's scope.*

# 7. Quality plan

## 7.1 Quality expectations

The deliverables must be correct, secure, traceable and reproducible by someone outside the team. Table 9 turns the Charter targets into measurable expectations; the numbers are targets to be validated, not results.

### Table 9: Quality targets and measures

| Quality attribute | Target | How it is measured | When / owner |
|---|---|---|---|
| **Answer accuracy** | ≥80% in English and in Vietnamese | Answers to the agreed bilingual test set graded Pass / Partial / Fail by two graders (Section 7.4). | Week 10 · Ny, Dang |
| **Retrieval quality (diagnostic)** | Report only | Share of questions where an expected source page is in the top-k results. | Week 10 · Dang |
| **Citation correctness** | 100% of answers cite a source or fall back | Check that each cited link opens the page that supports the claim. | Weeks 9–10 · Ny |
| **Response time** | <3 s average | End-to-end time from the Slack event to the posted reply, at least 30 queries after warm-up; report mean, median, 95th percentile, cached and uncached. | Week 10 · Dang, Khai |
| **Access control** | 100% whitelist compliance | Negative test matrix over channel, user and document; every non-permitted combination must be denied. | Week 10 · Nhat Hoang |
| **Code quality** | Every merge reviewed | Pull request with at least one reviewer, passing tests and secret scan; no direct pushes to main or develop. | Continuous · all |
| **Documentation quality** | Reproducible by VIB engineers | Two-pass review; a team member who did not write the section follows the deployment steps on the sandbox. | Weeks 11–12 · Hieu |
| **Cost** | Zero additional spend | Cost log reviewed at each sprint review. | Continuous · Hieu |

## 7.2 Quality assurance steps

- **Definition of done:** a backlog item is done when it is merged through a reviewed pull request, has tests where it contains logic (parser, chunker, whitelist, identity check), runs from the documented setup, and is recorded on the task board.
- **Reviews:** peer code review on every pull request; two-pass documentation check (content, then format and references); a sprint review with the sponsor every two weeks.
- **Secure by default:** deny-by-default whitelist, tokens kept in .env and never committed, secret scanning on every pull request, only mock or approved data.
- **Early evidence:** a vertical slice (Slack to security to RAG on mock data) is reviewed on 8 November, before the RAG integration, so that interface defects are found early.
- **Risk review:** the risk register is reviewed at the start of each weekly meeting and updated at the end of each sprint.

## 7.3 Planned testing

### Table 10: Test levels

| Level | Scope | Approach | Owner · week |
|---|---|---|---|
| **Unit** | HTML→Markdown converter, chunker, whitelist and identity middleware, prompt builder | Automated tests with sample pages and fixtures; parse-failure cases | Khai, Nhat Hoang · W7–9 |
| **Integration** | Slack → security → RAG → Slack; Confluence → index | Mock-backed contract tests, then runs in the Slack and Confluence sandboxes | Nhat Hoang, Minh Hoang · W8–9 |

| Level | Scope | Approach | Owner · week |
|---|---|---|---|
| **System (functional)** | Questions in English and Vietnamese, unanswerable questions, tables, images, restricted pages | Scripted scenarios in the sandbox; expected sources recorded in advance | Ny · W9–10 |
| **Accuracy benchmark** | Bilingual test set (Section 7.4) | Two-grader review, disagreements resolved by discussion | Ny, Dang · W10 |
| **Performance** | Latency under normal and repeated queries | Timed runs on stated hardware; cache on and off | Dang, Khai · W10 |
| **Security (negative)** | Non-whitelisted channels, unvalidated users, restricted documents, prompt-injection attempts | Test matrix with expected denials; results retained as evidence | Nhat Hoang, Minh Hoang · W10 |
| **Usability and acceptance** | Realistic queries reviewed by representative users; demonstration to the sponsor | Short review form (helpful, correct, clear); check-in on 12/11 and final demo | Hieu, Ny · W9–12 |

### 7.4 Accuracy evaluation method

The test set will contain at least 50 questions (proposed size, to be agreed with Mr. Phat on 29 October), split evenly between English and Vietnamese and covering prose, tables, images, restricted pages and questions that cannot be answered from the pages. Each question has an expected source page and a reference answer written before testing. Two team members grade each answer independently as **Pass** (correct, complete and correctly cited), **Partial** or **Fail**, then resolve differences by discussion. This follows the acceptance-checklist practice of Boonsri and Buranasaksee [2]. The primary accuracy figure is the share of answerable questions graded Pass; the normalised score (Pass plus half of Partial) is also reported. Unanswerable questions are scored on whether the fallback reply is returned. The definition of "accuracy" in the Charter is confirmed with the sponsor on 29 October.

### 7.5 Defects and exit criteria

Defects are logged as GitHub issues with a severity (critical, high, medium, low) and an owner. The Alpha release can be handed over when no critical or high defect is open, the accuracy, latency and security results are measured and written in the specification, the specification has been reviewed twice internally, and the sponsor's questions have been answered or recorded as known limitations. Requirement-to-test traceability is shown in Appendix C.

## 8. Research and investigations

The research below supported the requirements analysis and system design. Findings from each source are linked to a design decision; limits of each source are stated, so that the team does not over-generalise.

### 8.1 Application domain: enterprise knowledge and RAG

Retrieval-Augmented Generation combines a retriever that finds relevant passages with a generator that writes an answer from them, so that answers can point to their sources and can be updated by changing the documents instead of retraining a model [3]. This fits VIB's need to answer from frequently changing Confluence pages and to show where an answer comes from. In a bank, the retrieval step is also the place to enforce access: only documents that the asker may see should reach the model. The team therefore applies the whitelist and identity checks before retrieval (Part 2, Section 5).

### 8.2 Paper 1 – Chatbot System Architecture (Mohammed and Aref)

**Summary.** The paper describes a generic chatbot with three core components [1]: natural language understanding (topic detection, intent analysis, entity linking), a dialogue manager that chooses a response strategy (rule-based, knowledge-based, retrieval-based or generative, with context tracking), and a natural language generator (content filter and ranking of candidate replies). A case study of an

airline's messenger bot reports a 40% increase in interaction through the channel and a 15% share of online boarding bookings after launch.

**What we use.** The three-part pipeline is a clear way to explain and test our system. Table 11 maps it to our layers. The paper's priority order for strategies (templates first, then knowledge, then retrieval, then generation) supports our choice to answer only from retrieved content and to use a template fallback instead of free generation.

**Table 11: Generic chatbot architecture mapped to the project design**

| Generic component [1] | Our counterpart | Backlog / layer |
|---|---|---|
| **Natural language understanding (topic, intent, entities)** | Language detection (EN/VI), query clean-up, a simple check that the message is a knowledge question; named entities are handled by retrieval. | B9 · RAG core layer |
| **Dialogue manager: rule-based strategy** | Security rules run first: channel whitelist, identity check and fixed fallback templates. | B6, B11, B13 · Security layer |
| **Dialogue manager: knowledge / retrieval-based strategy** | Retriever over the Confluence index with top-k selection; the answer is built only from retrieved chunks. | B4, B7 · RAG core and ingestion layers |
| **Dialogue manager: generative strategy** | LLM used only to write the answer from retrieved text with mandatory citations; no open-ended chat. | B7, B8 · RAG core layer |
| **Context tracking and context record** | Not in Alpha; conversation memory is a Beta item. | B20 · Beta |
| **Natural language generator (content filter, ranking)** | Prompt rules and a citation check act as the content filter; Block Kit formats the reply in Slack. Engagement ranking is not needed for an enterprise Q&A bot. | B8, B14 · RAG and presentation layers |

**Limits.** The paper is a general overview. Its examples and sources date from before large language models (for instance, topic detection with older classifiers and tweets as an information source), it gives no accuracy or latency method, and the publication details are not shown in the copy supplied. We therefore use it for structure, not for technology choices or evaluation.

## 8.3 Paper 2 – A Repository-First Framework for Democratizing Chatbot Development (Boonsri and Buranasaksee)

**Summary.** The paper proposes a framework with four layers (presentation, application, domain, infrastructure) that depend only inward, a plugin mechanism for new use cases, an adapter interface for new channels and data sources, and an AI coding agent that generates code from short playbooks [2]. In a within-subjects experiment with nine developers and eight micro-tasks (144 runs), the framework with the agent finished tasks about 56% faster and at about 43% lower cost than the agent alone, needed fewer human interventions and had similar error rates; a separate 50-user study gave a mean usability (SUS) score of 70.6.

**What we use.** (1) The layering and adapter idea, applied to our Confluence source, vector store and LLM (risk R5, requirement NFR7). (2) The use of PostgreSQL with pgvector in the infrastructure layer, which supports keeping pgvector as an option next to Qdrant, which the authors also use for code indexing. (3) The pass / partial / fail acceptance checklist and normalised accuracy, which we adopt for our benchmark (Section 7.4). (4) The use of the System Usability Scale, which we may use after the pilot if the sponsor agrees.

**Limits.** The authors state that the study is small and under-powered for some effects, uses professional developers rather than novices, targets one stack (C# with LINE and Google Sheets) and tests domain-constrained scenarios. The reported figures are also not fully consistent between sections (for example the time ratio is given as 2.29, 2.44 and 2.48 in different places), so we treat them as indicative. The paper concerns how chatbots are built, not how a RAG assistant is evaluated, so it does not replace our own accuracy test.

## 8.4 Similar systems and alternatives

- **Hosted platforms and open frameworks.** The paper above [2] contrasts hosted platforms (Dialogflow, Lex, Watson) that are quick to set up but create lock-in, with open frameworks (Rasa, Botkit) that give control but need programming skill. Our zero-cost and bank-governance constraints favour an open, code-owned approach.
- **Commercial enterprise search.** Slack AI and Atlassian Rovo are out-of-the-box options. The team judged them too costly and too closed for this PoC (Part 2, Section 5.3); their pricing and features should be rechecked before any production decision.
- **Messenger bots in industry.** The airline case in [1] shows that a bot placed where users already work (Messenger) can raise interaction; this supports putting the assistant inside Slack rather than creating a new tool.

## 8.5 Technology investigation

- **Slack integration.** Slack's Bolt SDK for Python [5] handles events such as app_mention, and Socket Mode lets an app receive events over a WebSocket instead of a public HTTP endpoint [4]. This suits a laptop or sandbox PoC; for production at scale VIB may prefer the HTTP Events API, which the specification will describe. Rate limits and scopes (app_mentions:read, chat:write, channels:history) are checked in Week 7.
- **Confluence ingestion.** The Confluence Cloud REST API returns page bodies in several formats and exposes restrictions and attachments [6]. Macros and tables are the main parsing risk (R6), so the converter is tested on prose, table, image and restricted-page examples first.
- **Embeddings.** Multilingual models such as BGE-M3 [11] and multilingual-E5 [12] cover Vietnamese and English; Vietnamese-specific models such as PhoBERT [13] exist but are not designed for retrieval. The team will compare one multilingual option on the bilingual test set rather than assume performance.
- **Vector storage.** FAISS is a similarity-search library [7]; Qdrant is a vector database with payload filtering [8]; pgvector adds vector search to PostgreSQL [9]; Chroma is an embeddable open-source vector store [10]. Table 12 states the comparison criteria; the Week 7 decision record applies them.
- **LLM options.** A local open-source model (for example through Ollama) keeps data on the team's machine and costs nothing but is limited by hardware (R8); a hosted API or AWS Bedrock is faster to start but introduces cost, rate limits and cross-border data questions. The PoC uses mock data, so both are acceptable; the specification will explain the trade-off for VIB.
- **Evaluation.** Human grading of a fixed test set is used as the main method; automated "LLM-as-a-judge" grading is a possible helper but has known biases and is reviewed in a survey [14], so it is not used as the primary measure.

**Table 12: Vector store options and decision criteria (to be completed in the Week 7 decision record)**

| Option | Strengths for this project | Points to check | Role in decision |
|---|---|---|---|
| **FAISS [7]** | Fast, simple, runs in-process; no server | No built-in metadata filtering or access control; persistence and updates need extra code | Candidate for the smallest PoC |
| **Qdrant [8]** | Vector database with payload filters; runs in Docker; also used in [2] | Another service to run; memory use on student laptops | Candidate where metadata filtering is needed |
| **Chroma [10]** | Easy to embed in Python; open source | Maturity and filtering options to be checked | Candidate for rapid prototyping |
| **pgvector [9]** | SQL filters and joins fit access-control metadata; also appears in [2] | Needs PostgreSQL; query speed to be measured | Candidate if access metadata is stored in SQL |

### 8.6 Security and identity research

Slack sign-in identifies a user but does not by itself grant access to bank documents, so the design treats identity validation and document entitlements as separate steps and denies by default. The OWASP Top 10 for LLM applications [15] lists risks such as prompt injection and sensitive-information disclosure; the controls are to retrieve only permitted documents before generation, to keep the model from taking actions, and to test injection attempts in the security tests (Section 7.3).

### 8.7 Research responsibilities and open questions

Each member covers at least one research topic, recorded below. The open questions are the points the team could not settle from documents and will test in Weeks 7–8.

**Table 13: Research topics by member**

| Member | Topic | Used in |
|---|---|---|
| **Truong Ngoc Gia Hieu** | Enterprise-search alternatives (Slack AI, Atlassian Rovo), project governance and the specification outline | Section 5.3; Appendix B |
| **Pham Nguyen Minh Hoang** | Slack Bolt SDK, Socket Mode, scopes and rate limits | Backlog B5, B10; Section 8.5 |
| **Nguyen Quang Khai** | Confluence REST API and HTML/macro parsing | Backlog B1, B3, B12 |
| **Lu Nhat Hoang** | Access control, Slack identity and privacy law | Backlog B6, B11; Section 10 |
| **Duong Nguyen Dang** | Embeddings, vector stores and LLM options | Backlog B2, B4, B7; Table 12 |
| **Le Xuan Ny** | Evaluation method and bilingual test design | Section 7.4; backlog B15 |

- How well do the chosen embeddings retrieve Vietnamese content compared with English?
- Which Confluence page restrictions can be read through the API with the available token, and how are they mapped to Slack users?
- Does the free Slack workspace keep Socket Mode connected long enough for the demonstration, and what are the hosting needs at VIB?
- How large must the test set be for the 80% target to be meaningful? The proposed 50 questions is a minimum, not a statistical guarantee.

## 9. Skills analysis

The architecture requires skills across Slack interaction, access control, document ingestion, retrieval, testing, specification writing and delivery. The current roster records Artificial Intelligence majors for Dang and Ny and Software Development for Nhat Hoang; other majors and all detailed proficiency levels are not recorded. Table 1 therefore records evidence from task ownership and awaits confirmation by each member. Accordingly, the allocations below are a proposed capability-development plan based on current task ownership, not claims that members are already experts.

**Table 14: Skills analysis**

| Capability | Proposed owner / backup | Evidence and gap | Development plan and verification |
|---|---|---|---|
| **Slack / Python events** | Minh Hoang / Khai, Nhat Hoang | Task 27 ownership is recorded; live Socket Mode connectivity is unverified. | Run the starter; retain redacted evidence of allowed/blocked channels, thread routing and reconnection before acceptance. |
| **Security / identity / ACLs** | Nhat Hoang / Minh Hoang; Gia Hieu and client review | Requirements/ethics work provides context. Identity provider, access tiers and document entitlements are unresolved. | Week 7: agree a mock access matrix; deny unspecified access; test user, channel and document boundaries before retrieval. Slack sign-in is not bank authorization. |

| Capability | Proposed owner / backup | Evidence and gap | Development plan and verification |
|---|---|---|---|
| **Confluence ingestion** | Khai / Ny | Research/documentation assignments provide context. REST parsing, table extraction and OCR skills are unverified. | Week 7: ingest mock prose, table, image and restricted-page examples; compare extracted text and access metadata. Confirm attachment/OCR scope for Alpha. |
| **RAG / vector storage** | Dang / Ny | Both have recorded AI majors. Framework proficiency is unverified. Qdrant/FAISS options conflict with a newer pgvector proposal. | Week 7: record one minimal stack decision; Week 8: build a cited retrieval spike. Use a bilingual test set and expected sources. Measure the 80% accuracy and <3-second average targets on stated hardware in Week 10. |
| **Software / database interfaces** | Nhat Hoang / Khai | Software Development major and requirements work are recorded. SQL/vector-store and integration proficiency need evidence. | Define modular Slack-to-security-to-RAG contracts, add mock-backed interface tests, and review a working vertical slice on 8 November, before Week 9 integration. |
| **Quality / evaluation** | Ny / Minh Hoang (Slack), Dang (retrieval) | Quality Plan ownership is recorded. An agreed bilingual corpus and measured results are absent. | Define cases and criteria before integration (Week 9). Week 10: retain accuracy, latency, negative access results, sample sizes, configuration and failure analysis; Weeks 11–12: carry them into the specification test report. |
| **Technical specification writing** | Hieu, Nhat Hoang / Ny | Proposal and Charter sections show documentation ability; the team has not yet written a document that another team must deploy from. | Agree the outline with Mr. Phat on 12 November (Appendix B); draft sections during the build; two-pass review; a member outside the author group tests the deployment steps on the sandbox. |
| **Delivery / collaboration** | Gia Hieu / all members | The Charter confirms project management; repository tasks establish coordination. Individual Git/Python proficiency is unverified. | Each member runs setup and submits a small reviewed PR into develop. Share backups and a reproducible environment; the handover pack in Weeks 11–12 includes deployment steps and limitations. |

**Review mechanism:** each proposed owner demonstrates a small working artifact and identifies one backup who can reproduce it. Gia Hieu records observed capability, remaining gaps and adjusted ownership in Table 1 and the task board. Review weekly; if a member is unavailable or a spike fails, pair with the backup, reduce optional features and seek sponsor clarification. Keep mock-data development within the agreed zero-additional-spend constraint. Free services and Socket Mode do not guarantee continuous hosting.

## 10. Ethical considerations

The assistant reads the internal documents of a bank and uses a language model to write answers from them. The main ethical concerns are the confidentiality and privacy of information, accuracy and honesty of the answers, fairness between Vietnamese and English users, transparency about the use of AI, accountability, and compliance with the laws that apply to VIB, to an Australian university project and to the way the team works. During Project A only mock or approved low-sensitivity data is used, which keeps the privacy risk low, but the design and the specification must be ready for real data.

### 10.1 Applicable laws, principles and standards

**Table 15: Relevant instruments and the project response**

| Instrument | Why it matters | Project response |
|---|---|---|
| **Privacy Act 1988 (Cth) and the Australian Privacy Principles (APPs) [16][17]** | The project is carried out at an Australian university. APP 3 (collection), 5 (notice), 6 (use and disclosure), 8 (cross-border disclosure) and 11 (security) are the closest fit. | No personal information of real people is collected; data minimisation; no disclosure of personal information to an overseas LLM service; secrets and logs protected. For any real data VIB (as operator) decides the lawful basis. |
| **Australia's AI Ethics Principles [18]** | Eight voluntary principles: human, societal and environmental wellbeing; human-centred values; fairness; privacy protection and security; reliability and safety; transparency and explainability; contestability; accountability. | Answers are labelled as AI-generated and show sources (transparency); tests and fallbacks (reliability); a feedback route and a human owner for every answer path (contestability, accountability); bilingual parity checks (fairness). |

| Instrument | Why it matters | Project response |
|---|---|---|
| **Vietnam Law on Personal Data Protection, Law No. 91/2025/QH15, in force 1 January 2026, with Decree 356/2025/ND-CP [19]** | VIB staff are data subjects; consent, impact assessment and cross-border transfer rules apply to a real deployment. It carries over much of Decree 13/2023/ND-CP. | Mock data only in Project A. The specification lists the personal-data fields the system could touch and the assessments (processing and cross-border) that VIB must complete before using real data or an offshore model. |
| **Vietnam Law on Artificial Intelligence, passed 10 December 2025, in force 1 March 2026 [20]** | Risk-based rules, transparency duties and human oversight for AI systems; sector grace periods exist for existing systems, and details depend on Government decrees. | The assistant is an internal knowledge aid that makes no decisions about people. VIB compliance should confirm the risk class; the design keeps human oversight and discloses AI use. |
| **State Bank of Vietnam Circular 09/2020/TT-NHNN on information-system security in banking operations [21]** | Sets security expectations for banks' information systems (for example classification of information, access control and logging). | Deny-by-default access, no production data, audit logging planned (B24). VIB IT/security must confirm which requirements apply before deployment. |
| **OWASP Top 10 for LLM Applications [15]** | Names risks such as prompt injection and sensitive-information disclosure. | Retrieve only permitted documents before generation; no tool actions for the model; injection cases in the security tests. |
| **ACM Code of Ethics and Professional Conduct [22]** | Professional duties: avoid harm, be honest, respect privacy, give credit. | Honest reporting of results and limits; sources cited; respect of licences and platform terms. |
| **Platform and licence terms (Slack, Atlassian, model and library licences)** | Free tiers and open-source models carry usage and licence conditions. | Use sandboxes only for development; record each model and library licence in the specification. |

*This is a project-level review by students, not legal advice. Legal references were checked on the dates the sources were read; VIB legal and compliance staff must confirm which instruments apply to a real deployment.*

## 10.2 Ethical risks and controls

### Table 16: Ethical risks and controls

| Risk | Concern | Control | Owner |
|---|---|---|---|
| **Confidentiality / data leakage** | An answer exposes content that the asker may not see, or logs hold sensitive text. | Deny-by-default whitelist, identity check, permission filter before retrieval, redacted logs, no secrets in the repository, negative tests. | Nhat Hoang |
| **Hallucination and over-reliance** | Users take a wrong answer as an official policy. | Mandatory citations, fallback when no context is found, accuracy benchmark, notice that answers must be checked against the cited page. | Dang |
| **Language fairness** | Vietnamese users receive worse answers than English users. | Multilingual embeddings, per-language accuracy targets, bilingual test set. | Ny |
| **Transparency** | Users do not know they are talking to an AI. | Label responses as AI-generated; show the model and data source in the specification. | Minh Hoang |
| **Monitoring of staff** | Query logs reveal what individuals ask. | Collect only what is needed; no logging of queries in Alpha beyond debugging on mock data; audit logging (Beta) needs a notice, retention limit and access rules agreed with VIB. | Hieu |
| **Prompt injection** | A page or query tries to override instructions. | Instruction hierarchy in prompts, no tool actions, injection tests. | Nhat Hoang |
| **Intellectual property and licences** | Use of content or models beyond their licence. | Use only sandbox or approved content; record licences; cite sources. | Khai |

## 10.3 Data handling rules for the project

- Use mock or low-sensitivity content in the sandboxes until the Confluence administrators, Slack administrators and VIB IT/security approve other data in writing.
- Store tokens only in environment files; never commit them; scan every pull request for secrets.
- Do not place personal data in prompts, logs, test files or this repository; replace real names in examples.
- Share the indexed vector snapshot only inside the team; delete it when the project ends or when the sponsor asks.
- Record every external service used (Slack, Atlassian, model or API) and what data it receives.

### 10.4 Human oversight and transparency

The assistant answers only from retrieved pages and always shows the pages, so a person can verify the answer. When it cannot find a supported answer it says so instead of guessing. Owners of knowledge pages stay responsible for their content, and users can report wrong answers to the team during the pilot (a feedback button is a Beta item). No answer from the assistant triggers a transaction or an automated decision.

### 10.5 Use of generative AI in preparing this document

Generative AI tools were used to help draft and edit parts of this plan. The team has reviewed the text and is responsible for its content, and the use of such tools is declared in line with university academic-integrity requirements. [Team to confirm this declaration against the unit and university rules before submission.]

# References


[1] M. Mohammed and M. M. Aref, "Chatbot System Architecture," Faculty of Computer and Information Sciences, Ain Shams University, Cairo, Egypt, n.d. (paper supplied to the team; publication details not shown).

[2] R. Boonsri and U. Buranasaksee, "A Repository-First Framework for Democratizing Chatbot Development," in Proc. 2025 9th Int. Conf. Information Technology (InCIT), IEEE, 2025, pp. 775–782, doi: 10.1109/INCIT66780.2025.11276047.

[3] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in Advances in Neural Information Processing Systems 33 (NeurIPS 2020). arXiv:2005.11401.

[4] Slack, "Using Socket Mode," Slack developer documentation. https://docs.slack.dev/apis/events-api/using-socket-mode/

[5] Slack, "slack-bolt: The Bolt Framework for Python," Python Package Index. https://pypi.org/project/slack-bolt/

[6] Atlassian, "Confluence Cloud REST API," Atlassian developer documentation. https://developer.atlassian.com/cloud/confluence/rest/v2/intro/

[7] J. Johnson, M. Douze and H. Jégou, "Billion-scale similarity search with GPUs," IEEE Trans. Big Data, vol. 7, no. 3, pp. 535–547, 2021 (FAISS).

[8] Qdrant, "Qdrant documentation." https://qdrant.tech/documentation/

[9] pgvector, "Open-source vector similarity search for Postgres." https://github.com/pgvector/pgvector

[10] Chroma, "Chroma documentation." https://docs.trychroma.com/

[11] J. Chen et al., "BGE M3-Embedding: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation," arXiv:2402.03216, 2024.

[12] L. Wang et al., "Multilingual E5 Text Embeddings: A Technical Report," arXiv:2402.05672, 2024.

[13] D. Q. Nguyen and A. T. Nguyen, "PhoBERT: Pre-trained language models for Vietnamese," in Findings of the Association for Computational Linguistics: EMNLP 2020, pp. 1037–1042.

[14] J. Gu et al., "A Survey on LLM-as-a-Judge," arXiv:2411.15594, 2024.

[15] OWASP Foundation, "OWASP Top 10 for Large Language Model Applications." https://owasp.org/www-project-top-10-for-large-language-model-applications/

[16] Office of the Australian Information Commissioner (OAIC), "Australian Privacy Principles quick reference." https://oaic.gov.au/privacy/australian-privacy-principles/australian-privacy-principles-quick-reference

[17] OAIC, "APP Guidelines, Chapter 8: APP 8 – Cross-border disclosure of personal information." https://www.oaic.gov.au/privacy/australian-privacy-principles-guidelines/chapter-8-app-8-cross-border-disclosure-of-personal-information/

[18] Australian Government Department of Industry, Science and Resources, "Australia's AI Ethics Principles," 2019. https://www.industry.gov.au/publications/australias-artificial-intelligence-ethics-framework/australias-ai-ethics-principles

[19] Tilleke & Gibbins, "Vietnam's New Personal Data Protection Law: A Closer Look." https://tilleke.com/insights/vietnams-new-personal-data-protection-law-a-closer-look

[20] Baker McKenzie, "Vietnam: Artificial Intelligence Law – foundation and outlook," Feb. 2026. https://www.bakermckenzie.com/en/insight/publications/2026/02/vietnam-artificial-intelligence-law-foundation-and-outlook

[21] State Bank of Vietnam, Circular No. 09/2020/TT-NHNN of 21 October 2020 on the security of information systems in banking operations (English version). https://english.luatvietnam.vn/circular-no-09-2020-tt-nhnn-dated-october-21-2020-of-the-state-bank-of-vietnam-promulgating-the-security-of-information-system-in-banking-operations-192825-Doc1.html

[22] Association for Computing Machinery, "ACM Code of Ethics and Professional Conduct," 2018. https://www.acm.org/code-of-ethics

# Appendices

Appendix A gives the revised work breakdown, Appendix B the proposed outline of the Technical Specification Document, Appendix C the traceability matrix, Appendix D a snapshot of the task board and Appendix E a glossary.

# Appendix A. Revised projected schedule (work breakdown)

Dates are in 2026 (dd/mm). Phases I and II show the status recorded on the task board; Phases III to V were revised on 25 October 2026. Owners are shown as primary / backup. The same plan, with comments, is in the workbook sheet Project_Scheduled.

### Table 17: Work breakdown structure

| WBS | Task | Owner | Start | End | Days | Status |
|---|---|---|---|---|---|---|
| **1.0** | **Phase I: Initiation, Proposal Iteration and Charter Alignment** | **Team** | **14/09** | **18/10** | **35** | **Complete** |
| 1.1 | Requirements Exploration and Architecture Research | Team | 14/09 | 27/09 | 14 | Complete |
| 1.2 | Initial Discussion and Scoping Session | Team | 14/09 | 20/09 | 7 | Complete |
| 1.3 | Research RAG Architecture and Vector DB Options | Team | 21/09 | 27/09 | 7 | Complete |
| 1.4 | Project Proposal Drafting, Revision and Submission | Team | 28/09 | 18/10 | 21 | Complete |
| 1.5 | Project Charter Formalization & Structuring | Team | 02/10 | 18/10 | 17 | Complete |
| 1.6 | Final Review and Submission to Mr. Phat | Hieu | 14/10 | 18/10 | 5 | Complete |
| **2.0** | **Phase II: System Architecture Design & Environment Setup** | **Team** | **19/10** | **25/10** | **7** | **In Progress** |
| 2.1 | Repository Initialization & Team Onboarding | Team | 12/10 | 18/10 | 7 | Complete |
| 2.2 | Write Project_Scheduled (Task 20) | Nhat Hoang | 20/10 | 20/10 | 1 | Complete |
| 2.3 | Design Layered Architecture (Task 21) | Team | 19/10 | 25/10 | 7 | In Progress |
| 2.4 | Tech Stack, DWH & LLM Decision Matrix | Team | 19/10 | 25/10 | 7 | In Progress |
| 2.5 | Confluence Data Ingestion & REST Pipeline Strategy | Team | 19/10 | 25/10 | 7 | In Progress |
| 2.6 | Security Requirements Clearance & Client Questions | Gia Hieu | 20/10 | 25/10 | 6 | In Progress |
| 2.7 | Slack & Confluence Sandbox & GitHub Branch Setup | Hieu / Team | 22/10 | 23/10 | 2 | Complete |
| 2.8 | Project Document Writing (Sections 27–45) | Team | 23/10 | 24/10 | 2 | In Progress |
| **3.0** | **Phase III: Delivery Phase 1 - Data Foundation, Slack Gateway & Security Middleware** | **Team** | **26/10** | **08/11** | **14** | **Not Started** |
| 3.1 | Confluence REST API connector - fetch pages from mock/approved spaces (B1) | Khai / Ny | 26/10 | 30/10 | 5 | Not Started |
| 3.2 | Local vector DB, Docker dev environment & stack decision record (B2) | Dang / Ny | 26/10 | 30/10 | 5 | Not Started |
| 3.3 | Slack Bolt + Socket Mode bot: @mention -> threaded reply (B5) | Minh Hoang / Khai | 26/10 | 30/10 | 5 | Not Started |
| 3.4 | Channel whitelist middleware (deny by default) with negative tests (B6) | Nhat Hoang / Minh Hoang | 28/10 | 01/11 | 5 | Not Started |
| 3.5 | Mock knowledge base in VIB-Mock-Knowledge: EN/VI prose, tables, images, restricted pages | Ny / Khai | 26/10 | 01/11 | 7 | Not Started |
| 3.6 | Sponsor check-in #1 - confirm test data, Confluence access and Technical Specification expectations | Hieu | 29/10 | 29/10 | 1 | Not Started |
| 3.7 | HTML -> clean Markdown converter for tables and macros, with parse-failure log (B3) | Khai / Ny | 02/11 | 06/11 | 5 | Not Started |
| 3.8 | Chunking, multilingual (EN/VI) embedding and indexing pipeline (B4) | Dang / Ny | 04/11 | 08/11 | 5 | Not Started |
| 3.9 | Slack SSO / identity validation middleware (B11) | Nhat Hoang / Minh Hoang | 02/11 | 06/11 | 5 | Not Started |
| 3.10 | Interface contracts (Slack -> Security -> RAG) and mock-backed integration tests | Nhat Hoang / Khai | 05/11 | 08/11 | 4 | Not Started |
| 3.11 | Milestone M1: Phase 1 exit review (ingestion + gated bot demo) | Hieu / Team | 08/11 | 08/11 | 1 | Not Started |
| **4.0** | **Phase IV: Delivery Phase 2 - RAG Core, Slack Integration & Validation** | **Team** | **09/11** | **22/11** | **14** | **Not Started** |
| 4.1 | Retriever and LLM integration - basic Q&A (B7) | Dang / Ny | 09/11 | 13/11 | 5 | Not Started |
| 4.2 | Prompt templates with mandatory source citations and EN/VI query handling (B8, B9) | Dang / Ny | 11/11 | 15/11 | 5 | Not Started |

| WBS | Task | Owner | Start | End | Days | Status |
|---|---|---|---|---|---|---|
| 4.3 | Slack message event handling linked to the RAG backend (B10) | Minh Hoang / Nhat Hoang | 11/11 | 15/11 | 5 | Not Started |
| 4.4 | Embedded image asset handling (B12) | Khai / Ny | 11/11 | 15/11 | 5 | Not Started |
| 4.5 | Bilingual test query set with expected sources (preparation for B15) | Ny / Dang | 09/11 | 15/11 | 7 | Not Started |
| 4.6 | Sponsor check-in #2 - Phase 1 demo and Technical Specification outline approval | Hieu | 12/11 | 12/11 | 1 | Not Started |
| 4.7 | Fallback reply for unanswerable queries (B13) and Block Kit answers with clickable sources (B14) | Minh Hoang / Dang | 16/11 | 19/11 | 4 | Not Started |
| 4.8 | Accuracy benchmark >=80% on the bilingual test set (B15) | Ny / Dang | 18/11 | 22/11 | 5 | Not Started |
| 4.9 | Latency benchmark <3 s and response caching (B16) | Dang / Khai | 18/11 | 22/11 | 5 | Not Started |
| 4.10 | Security boundary tests - 100% whitelist compliance (B17) | Nhat Hoang / Minh Hoang | 18/11 | 22/11 | 5 | Not Started |
| 4.11 | Technical Specification skeleton and component sections drafted as components land | Hieu / Nhat Hoang | 09/11 | 22/11 | 14 | Not Started |
| 4.12 | Bug fixing, refactoring and Alpha build tag | Team | 20/11 | 22/11 | 3 | Not Started |
| 4.13 | Milestone M2: Alpha code freeze (Phase 2 exit) | Hieu / Team | 22/11 | 22/11 | 1 | Not Started |
| **5.0** | **Phase V: Technical Specification Document & Handover to Mr. Phat** | **Team** | **23/11** | **06/12** | **14** | **Not Started** |
| 5.1 | Spec Part A - Architecture, data flow and interface contracts | Nhat Hoang / Minh Hoang | 23/11 | 27/11 | 5 | Not Started |
| 5.2 | Spec Part B - Ingestion and RAG design: schema, models, chunking, prompts, configuration | Khai / Dang | 23/11 | 27/11 | 5 | Not Started |
| 5.3 | Spec Part C - Security, access control and privacy controls (whitelist, SSO, secrets, ethics) | Nhat Hoang / Ny | 24/11 | 28/11 | 5 | Not Started |
| 5.4 | Spec Part D - Test and validation report: accuracy, latency, security results, known limitations | Ny / Dang | 24/11 | 29/11 | 6 | Not Started |
| 5.5 | Spec Part E - Deployment and operations guide for the VIB environment | Minh Hoang / Khai | 24/11 | 29/11 | 6 | Not Started |
| 5.6 | Sponsor check-in #3 - walkthrough of the specification draft | Hieu | 26/11 | 26/11 | 1 | Not Started |
| 5.7 | Two-pass internal review and consolidation -> Technical Specification v1.0 | Hieu / Team | 30/11 | 01/12 | 2 | Not Started |
| 5.8 | Milestone M3: Send Technical Specification Document to Mr. Phat | Hieu | 02/12 | 02/12 | 1 | Not Started |
| 5.9 | Sponsor review, Q&A and specification revisions (v1.1) | Team | 02/12 | 05/12 | 4 | Not Started |
| 5.10 | Final demo, demo script and handover pack (repo tag, README, test evidence) | Hieu / Team | 04/12 | 06/12 | 3 | Not Started |
| 5.11 | Milestone M4: Final handover and Project A submission | Hieu | 06/12 | 06/12 | 1 | Not Started |
| 5.12 | VIB deployment from the specification (outside team scope; timing set by VIB) | Mr. Phat / VIB | | | | Not Started |

# Appendix B. Proposed outline of the Technical Specification Document

The specification is written for VIB engineers who deploy and maintain the assistant without the team present. The outline below is proposed for approval by Mr. Phat at the 12 November check-in. Drafting starts in Week 9 as components are completed; consolidation, review and the sponsor's questions fill Weeks 11–12.

**Table 18: Outline of the Technical Specification Document**

| No | Section | Content | Lead | Draft |
|---|---|---|---|---|
| 1 | Introduction | Purpose, audience, scope, how to read the document, version history. | Hieu | W9–10 |
| 2 | System overview | Context diagram, goals, assumptions, constraints, glossary. | Hieu, Nhat Hoang | W9–10 |
| 3 | Architecture | Four layers, components, data flow, interface contracts between Slack, security and RAG, decision records. | Nhat Hoang, Minh Hoang | W10–11 |
| 4 | Ingestion and storage | Confluence connector, parsing rules for tables, macros and images, chunking, embedding model, vector store schema, re-indexing. | Khai, Dang | W10–11 |
| 5 | RAG engine | Retrieval settings, prompt templates, citation rules, language handling, fallback, caching. | Dang | W10–11 |
| 6 | Slack application | App manifest, scopes, Socket Mode and HTTP Events options, event handling, Block Kit format. | Minh Hoang | W10–11 |
| 7 | Security and privacy | Whitelist, identity validation, access matrix, secrets, logging, legal and ethical controls, decisions VIB must take. | Nhat Hoang, Ny | W11 |
| 8 | Configuration reference | Environment variables, configuration files, defaults, examples. | Khai, Minh Hoang | W11 |
| 9 | Deployment and operations guide | Prerequisites, step-by-step setup, hosting options, monitoring, backup, runbook for common failures. | Minh Hoang, Khai | W11 |
| 10 | Test and validation report | Test set, accuracy, latency and security results, sample sizes, configuration used, failure analysis. | Ny, Dang | W10–11 |
| 11 | Known limitations and roadmap | Open issues, risks, Beta items (B19–B26) and recommended next steps. | Hieu | W11–12 |
| 12 | Appendices | Repository structure, dependency and licence list, change log, review and acceptance sign-off. | Dang, Hieu | W12 |

# Appendix C. Requirements-to-test traceability matrix

### Table 19: Traceability matrix

| Require ment | Backlog | Test / evidence | Pass condition | Planned |
|---|---|---|---|---|
| FR1 | B1 | Integration run against VIB-Mock-Knowledge; page metadata check | All agreed pages fetched with title, URL, date, restrictions | W7 |
| FR2 | B3 | Unit tests on prose, table, macro and broken pages; parse-failure log | Tables and macros converted; failures logged, not silent | W8 |
| FR3 | B12 | Image pages in the mock set | Images linked to pages; OCR scope as agreed | W9 |
| FR4 | B2, B4 | Index build and count check; EN and VI sample queries | Every chunk indexed with source metadata | W8 |
| FR5 | B5, B10 | Sandbox run: @mention in a whitelisted channel; reconnection test | Threaded reply received; reconnects after drop | W7, W9 |
| FR6 | B7 | Spike with expected sources | Relevant chunks retrieved and used in the answer | W9 |
| FR7 | B8, B14 | Citation check on every benchmark answer | 100% of answers show a correct source link or fallback | W10 |
| FR8 | B9 | Mixed EN/VI query set | Reply language equals question language | W9 |
| FR9 | B13 | Unanswerable questions in the test set | Fallback returned, no invented answer | W10 |
| FR10 | B6, B17 | Negative matrix: non-whitelisted channels | 100% denied | W7, W10 |
| FR11 | B11, B17 | Negative matrix: unvalidated users | 100% denied | W8, W10 |
| FR12 | B14 | Visual review of Block Kit output | Readable answer with source links | W10 |
| FR13 | B16 | Cached versus uncached timing | Repeated query served from cache | W10 |
| FR14 | B5, B6 | Change settings without editing code | Whitelist and model changed through configuration only | W7 |
| NFR1 | B15 | Accuracy benchmark (Section 7.4) | ≥80% per language | W10 |
| NFR2 | B16 | Latency benchmark | Average <3 s | W10 |
| NFR3 | B6, B11, B17 | Negative tests; secret scan on pull requests | 100% compliance; no secrets committed | W7–10 |
| NFR4 | All | Data-handling checklist at sprint reviews | Only mock or approved data found | Each sprint |
| NFR5 | B9, B15 | Per-language results | Both languages meet target | W10 |
| NFR6 | B5, B13 | Failure-injection tests (LLM or database down) | Fallback or clear error, no crash | W10 |
| NFR7 | B2, B4 | Interface contracts and mock-backed tests | Source, store and LLM replaced by a mock without code change in other layers | W8 |
| NFR8 | All | Pull request review; specification checklist | Every merge reviewed; documented setup works | Continuous |
| NFR9 | All | Cost log | Zero additional spend | Each sprint |
| NFR10 | B14 | Representative-user review form | Answers rated clear and helpful | W10–12 |
| NFR11 | B18 | This matrix reviewed with the sponsor | Every requirement has a test and evidence | W11–12 |

# Appendix D. Task board snapshot (Weekly Tasks, Weeks 5–6)

Snapshot of the sheet Weekly Tasks in "Computing Technology Project A.xlsx" on 25 October 2026, shown as evidence of the shared project management board. The status "Not Yet" marks sections that were still open before this draft was written; they are drafted in this document and await team review.

### Table 20: Task board snapshot

| No | Task | Owner | Start | Due | Status |
|---|---|---|---|---|---|
| | **Week05 (Oct 12th - 18th)** | | | | |
| | **Document: Project Proposal** | | | | |
| 1 | Write SUMMARY section | Hieu | 11/10 | 12/10 | Completed |
| 2 | Write DESIRED OUTCOME section | Ny | 11/10 | 12/10 | Completed |
| 3 | Write ACTION TO COMPLETE section | Khai | 11/10 | 12/10 | Completed |
| 4 | Write BENEFITS OF PROJECT section | Minh Hoang | 11/10 | 12/10 | Completed |
| 5 | Write PROJECTED SCHEDULE section | Nhat Hoang | 11/10 | 12/10 | Completed |
| 6 | Write PROJECTED BUDGET section | Dang | 11/10 | 12/10 | Completed |
| 7 | Write PROJECTED TEAM AND RESOURCE REQUIREMENTS section | Hieu | 11/10 | 12/10 | Completed |
| | **Document: Project Charter** | | | | |
| 8 | Write PROBLEM OR ISSUE section | Ny | 14/10 | 16/10 | Completed |
| 9 | Write PURPOSE OF PROJECT section | Khai | 14/10 | 16/10 | Completed |
| 10 | Write BUSINESS CASE section | Minh Hoang | 14/10 | 16/10 | Completed |
| 11 | Write GOALS/METRICS section | Nhat Hoang | 14/10 | 16/10 | Completed |
| 12 | Write EXPECTED DELIVERABLES section | Dang | 14/10 | 16/10 | Completed |
| 13 | Write WITHIN SCOPE section | Hieu | 14/10 | 16/10 | Completed |
| 14 | Write OUTSIDE OF SCOPE section | Ny | 14/10 | 16/10 | Completed |
| 15 | Write TENTATIVE SCHEDULE section | Khai | 14/10 | 16/10 | Completed |
| 16 | Write RESOURCES section | Minh Hoang | 14/10 | 16/10 | Completed |
| 17 | Write COSTS section | Nhat Hoang | 14/10 | 16/10 | Completed |
| 18 | Write BENEFITS AND CUSTOMERS section | Dang | 14/10 | 16/10 | Completed |
| 19 | Write RISKS, CONSTRAINTS, and ASSUMPTIONS section | Hieu | 14/10 | 16/10 | Completed |
| | **Week06 (Oct 19th - 25th)** | | | | |
| 20 | Write Project_Scheduled | Nhat Hoang | 22/10 | 23/10 | Completed |
| 21 | Write Announcement Email | Hieu | 22/10 | 23/10 | Completed |
| 22 | Write Announcement Email | Hieu | 22/10 | 23/10 | Completed |
| 23 | Enter Slack and Confluence Email | Hieu | 22/10 | 23/10 | Completed |
| 24 | Add Github User | Hieu | 22/10 | 23/10 | Completed |
| 25 | Take Note Meeting Minutes (MoM) | Ny | 22/10 | 23/10 | In Progress |
| 26 | Make Slide for Presentation | Dang | 22/10 | 23/10 | Completed |
| 27 | Figure out how to Set up Framework for Bot in Slack and then choose the best one later | Minh Hoang | 22/10 | 24/10 | In Progress |
| 28 | Set Up Branches for Github | Hieu | 22/10 | 23/10 | Completed |
| 29 | Edit Cover Page section | Hieu | 23/10 | 24/10 | Completed |
| 30 | Write Acknowledgement of Country section | Hieu | 23/10 | 24/10 | Completed |
| 31 | Write Team Profile (Table 1) section | Khai | 23/10 | 24/10 | Not Yet |
| 32 | Write Team Role (Table 2) section | Minh Hoang | 23/10 | 24/10 | Completed |
| 33 | Write TeamWork Roadmap section | Nhat Hoang | 23/10 | 24/10 | In Progress |
| 34 | Write Document Management section | Dang | 23/10 | 24/10 | Completed |

| No | Task | Owner | Start | Due | Status |
|---|---|---|---|---|---|
| 35 | Write Risk Mitigation section | Hieu | 23/10 | 24/10 | Completed |
| 36 | Write Problem Statetment section | Ny | 23/10 | 24/10 | Not Yet |
| 37 | Write Scope section | Khai | 23/10 | 24/10 | Not Yet |
| 38 | Write Stakeholder section | Minh Hoang | 23/10 | 24/10 | Completed |
| 39 | Write High-Levrl Requirements section | Nhat Hoang | 23/10 | 24/10 | Completed |
| 40 | Write Solution Approach section | Dang | 23/10 | 24/10 | Completed |
| 41 | Write Product Backlog section | Hieu | 23/10 | 24/10 | Completed |
| 42 | Write Quality Plan section | Ny | 23/10 | 24/10 | Not Yet |
| 43 | Write Research and Investigation section | Khai | 23/10 | 24/10 | Not Yet |
| 44 | Write Skills Analysis section | Minh Hoang | 23/10 | 24/10 | Completed |
| 45 | Write Ethical Considerations section | Nhat Hoang | 23/10 | 24/10 | Completed |
| 46 | Attach File Project Proposal and Project Charter to the Confluence workspace. | Ny | 23/10 | 24/10 | Not Yet |

# Appendix E. Glossary

### Table 21: Glossary and abbreviations

| Term | Meaning |
|---|---|
| **Alpha / Beta** | Releases of the project: Alpha is the proof of concept built in Project A (Weeks 5–12); Beta extends it in Project B. |
| **Block Kit** | Slack's format for rich messages such as sections, buttons and links. |
| **Bolt SDK** | Slack's software kit for building Slack apps (here in Python). |
| **Chunking** | Splitting a document into smaller passages before they are embedded and indexed. |
| **Confluence** | Atlassian's wiki product where VIB keeps internal documentation. |
| **Embedding** | A list of numbers that represents the meaning of a text so similar texts can be found. |
| **LLM** | Large language model, a model that writes text; here used to write answers from retrieved passages. |
| **PoC** | Proof of concept, a working prototype that tests whether an idea is feasible. |
| **RAG** | Retrieval-Augmented Generation: find relevant passages first, then write an answer from them. |
| **RBAC / ACL** | Role-based access control / access control list: rules about who may see what. |
| **Socket Mode** | A way for a Slack app to receive events over a WebSocket connection instead of a public web address. |
| **SSO** | Single sign-on: users prove who they are once through the company identity system. |
| **SUS** | System Usability Scale, a ten-question usability questionnaire. |
| **Vector database** | A database that stores embeddings and returns the nearest ones to a query. |
| **WBS** | Work breakdown structure: the hierarchical list of project tasks. |
| **p95** | The 95th percentile: 95% of measurements are at or below this value. |